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PROJECT REPORT

**Pilot-Aided Channel Estimation and
Pilot-Density Trade-offs in
Time-Varying MIMO-OFDM Systems**



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ABSTRACT

This report presents a MATLAB physical-layer simulator for a coded spatial-multiplexing MIMO-OFDM link. It studies the tradeoff between pilot overhead and channel tracking in a time-varying frequency-selective multipath channel. The simulator includes QAM, a rate-one-half convolutional code with soft Viterbi decoding, orthogonal pilot symbols, Rayleigh tapped-delay-line fading with Jakes-inspired temporal correlation, least-squares and DFT-filtered channel estimation, and ZF/MMSE equalization. BER, RMS EVM, channel-estimation NMSE, and normalized payload efficiency demonstrate that frequent pilots improve channel tracking and reliability while reducing payload resources. The report documents the actual model and explicitly states its limits.

Keywords: MIMO-OFDM, Channel Estimation

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Chapter 1

Introduction

1.1 Motivation

Wireless propagation creates delayed reflections from objects in the environment. These copies add with different phase and amplitude, producing frequency-selective fading. Mobility causes the channel to evolve with time. Orthogonal frequency-division multiplexing (OFDM) addresses frequency selectivity by transmitting many narrowband subcarriers in parallel, while multiple-input multiple-output (MIMO) transmission can send several spatial streams simultaneously. The receiver must estimate the channel accurately enough to separate those streams.

The central question of this project is: **“How do pilot periodicity and DFT-domain channel-estimation filtering affect reliability and payload efficiency in time-varying MIMO-OFDM?”**

1.2 Contributions

The project implements configurable $N_{Tx} \times N_{Rx}$ spatial multiplexing, QPSK/16-QAM/64-QAM, dedicated full-band orthogonal pilots, a time-correlated Rayleigh tapped-delay-line channel, LS and DFT-linear estimators, ZF/MMSE equalization, convolutional coding, soft Viterbi decoding, and reproducible Monte-Carlo studies.

Chapter 2

Review of Literature

Modern broadband wireless systems combine orthogonal frequency-division multiplexing (OFDM), multiple-input multiple-output (MIMO) transmission, and pilot-assisted channel estimation to achieve reliable communication over frequency-selective fading channels. The foundations of OFDM were established by Weinstein and Ebert [1], while Telatar [2] demonstrated the substantial capacity gains achievable through multiple antennas. Practical aspects of MIMO communication were later consolidated by Paulraj *et al.* [3]. The underlying principles of wireless propagation, digital communication, and information theory are comprehensively covered in the standard texts by Tse and Viswanath [4], Goldsmith [5], Proakis and Salehi [6], and Stüber [7].

Reliable coherent detection depends on accurate channel estimation. Among the available techniques, pilot-assisted estimation remains the most widely adopted because of its simplicity and practical implementation. Coleri *et al.* [8] demonstrated the influence of pilot arrangement and interpolation strategy on estimation performance in OFDM systems. To improve the performance of conventional least-squares (LS) estimation, transform-domain approaches have been proposed that exploit the finite delay spread of practical wireless channels. Edfors *et al.* [9] introduced an SVD-based estimator, while Yeh and Wilson [10] showed that DFT-based delay-domain truncation effectively suppresses estimation noise without increasing pilot overhead. At the receiver, linear MIMO detectors remain attractive because of their favourable complexity-performance trade-off, with MMSE equalization generally preferred over Zero-Forcing owing to its improved robustness against noise [3, 11].

Modern wireless standards, including 5G New Radio, integrate OFDM, MIMO, pilot-based reference signals, and advanced channel coding to support reliable communication under time-varying propagation conditions [12]. Although pilot-assisted channel estimation is well established, selecting an appropriate pilot density remains an important design trade-off: frequent pilots improve channel tracking but reduce spectral efficiency, whereas sparse pilots increase payload transmission at the expense of estimation accuracy. Motivated by this trade-off, the present work develops a coded MIMO-OFDM simulation framework with time-varying Rayleigh fading, LS and DFT-enhanced channel estimation, MMSE equalization, and soft-decision decoding to systematically evaluate the impact of pilot density on reliability, estimation accuracy, and payload efficiency.

Chapter 3

Theory and Background

3.1 OFDM waveform

For FFT size N , the frequency-domain values $X_m[k]$ are transformed into a time-domain OFDM symbol:

$$x_m[n] = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} X_m[k] e^{j2\pi kn/N}. \quad (3.1)$$

A cyclic prefix copies the final N_{CP} samples to the front. If the channel has L taps and

$$N_{\text{CP}} \geq L - 1, \quad (3.2)$$

then CP removal followed by an FFT produces the simple frequency-domain relation

$$Y_m[k] = H_m[k]X_m[k] + W_m[k]. \quad (3.3)$$

The research simulator calculates this post-FFT equivalent relation directly. Under the CP and block-fading assumptions, it is mathematically equivalent to explicit IFFT/CP waveform simulation, while allowing faster Monte-Carlo sweeps.

3.2 MIMO spatial multiplexing

At subcarrier k and OFDM symbol m :

$$\mathbf{y}_m[k] = \mathbf{H}_m[k]\mathbf{s}_m[k] + \mathbf{w}_m[k], \quad (3.4)$$

where $\mathbf{H}_m[k] \in \mathbb{C}^{N_{\text{Rx}} \times N_s}$ is the MIMO channel, $\mathbf{s}_m[k]$ contains N_s simultaneous QAM streams, and $\mathbf{w}_m[k] \sim \mathcal{CN}(\mathbf{0}, N_0 \mathbf{I})$. The baseline configuration is 2×2 MIMO with two streams. Each receive antenna observes a mixture of both transmit streams; the receiver uses an estimate of $\mathbf{H}_m[k]$ to separate them.

3.3 Multipath and Doppler physics

The channel has L complex delayed taps. Its frequency response is

$$H_{m,r,t}[k] = \sum_{\ell=0}^{L-1} h_{m,r,t}[\ell] e^{-j2\pi k\ell/N}. \quad (3.5)$$

The implementation evolves every tap once per OFDM symbol:

$$h_{m+1}[\ell] = \rho h_m[\ell] + \sqrt{1 - \rho^2} u_m[\ell], \quad \rho = J_0(2\pi f_D T_{\text{sym}}), \quad (3.6)$$

where $u_m[\ell]$ is complex Gaussian innovation and $f_D T_{\text{sym}}$ is normalized Doppler. Larger Doppler means faster decorrelation and more rapidly stale pilot estimates. This symbol-wise model captures channel aging, but not channel variation within one OFDM symbol or full Doppler ICI.

3.4 Modulation and coding

Gray-labelled QPSK, 16-QAM, and 64-QAM are normalized to unit average symbol energy. The baseline uses QPSK:

$$s = \frac{(1 - 2b_I) + j(1 - 2b_Q)}{\sqrt{2}}. \quad (3.7)$$

Information bits are encoded by a rate-1/2, constraint-length-3 convolutional code with generators $(7, 5)_8$. At the receiver, max-log bit log-likelihood ratios are computed:

$$L(b) \approx \frac{\min_{a:b(a)=1} |z - a|^2 - \min_{a:b(a)=0} |z - a|^2}{N_0}, \quad (3.8)$$

and a soft Viterbi decoder estimates the most likely information-bit sequence.

Chapter 4

System Design

This chapter describes the architecture of the developed MIMO–OFDM communication system. Figure 4.1 illustrates the complete baseband signal processing chain implemented in the MATLAB simulator.

The transmitter begins by encoding the information bits using a rate-1/2 convolutional encoder before mapping them onto Gray-labelled QAM symbols. Orthogonal pilot symbols are periodically inserted into the transmitted frame to facilitate channel estimation. Following OFDM modulation, the signal propagates through a time-varying frequency-selective MIMO Rayleigh fading channel. At the receiver, OFDM demodulation is followed by pilot-assisted channel estimation, linear MIMO equalization, soft demodulation, and Viterbi decoding to recover the transmitted information sequence.

4.1 Frame Structure and Pilot Arrangement

The transmitted frame consists of periodically inserted pilot events interleaved with payload OFDM symbols. During a pilot event, only one transmit antenna radiates a known full-band pilot sequence while all remaining antennas remain silent. Consequently,

$$X_{p_t}[k, t] = 1, \quad X_{p_t}[k, t'] = 0, \quad t' \neq t. \quad (4.1)$$

For a two-transmit-antenna configuration, each pilot event occupies two OFDM symbols so that the channels associated with both transmit antennas can be estimated independently.

A pilot period P denotes the interval between successive pilot events. The corresponding pilot overhead is approximately

$$\eta_{\text{pilot}} = \frac{N_{\text{Tx}}}{P}. \quad (4.2)$$

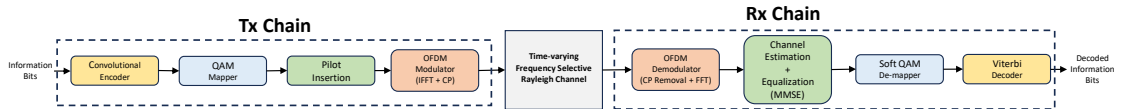


Figure 4.1: Block diagram of the coded MIMO-OFDM transceiver considered in this work.

Smaller pilot periods improve channel tracking accuracy but reduce the fraction of OFDM symbols available for payload transmission. The investigated pilot configurations are illustrated in the following section.

4.2 Channel Estimation

At pilot locations, the receiver estimates the frequency-domain channel coefficients using the least-squares (LS) estimator,

$$\hat{H}_{p,r,t}^{\text{LS}}[k] = \frac{Y_p[k, r]}{X_p[k, t]}. \quad (4.3)$$

The estimated pilot responses are linearly interpolated across neighbouring data symbols to obtain channel estimates throughout the frame. To exploit the finite delay spread of practical multipath channels, the LS estimate is further refined using DFT-domain processing,

$$\tilde{h} = \text{IFFT} \left\{ \hat{H}^{\text{LS}} \right\}, \quad (4.4)$$

$$\tilde{h}[\ell] = 0, \quad \ell \geq \hat{L}, \quad (4.5)$$

$$\hat{H}^{\text{DFT}} = \text{FFT} \left\{ \tilde{h} \right\}. \quad (4.6)$$

Delay-domain truncation suppresses noise components that lie outside the physically expected delay support, thereby improving channel estimation accuracy without increasing pilot overhead. The effectiveness of this approach depends on selecting an appropriate delay-support threshold $\hat{L}$.

4.3 MIMO Equalization

The equalizer reconstructs the transmitted spatial streams using the estimated channel matrix. The Zero-Forcing detector is given by

$$\mathbf{W}_{\text{ZF}} = \text{pinv} \left(\hat{\mathbf{H}} \right), \quad \hat{\mathbf{s}} = \mathbf{W}_{\text{ZF}} \mathbf{y}. \quad (4.7)$$

Although the ZF detector effectively suppresses inter-stream interference, it may significantly amplify receiver noise when the channel matrix is poorly conditioned. To improve robustness, the simulator primarily employs Minimum Mean-Square Error (MMSE) equalization,

$$\mathbf{W}_{\text{MMSE}} = \left(\hat{\mathbf{H}}^H \hat{\mathbf{H}} + N_0 \mathbf{I} \right)^{-1} \hat{\mathbf{H}}^H. \quad (4.8)$$

By regularizing the matrix inversion, MMSE equalization provides a better compromise between interference suppression and noise enhancement, resulting in superior performance under practical signal-to-noise ratios.

Chapter 5

Methodology

Table 5.1: Baseline simulation configuration.

Parameter	Value
FFT size / CP length	64 / 16 samples
Channel length	8 taps
Antennas and streams	2×2 , two streams
Frame length	80 OFDM symbols
Modulation and coding	QPSK; rate-1/2 convolutional code
Estimator / equalizer	DFT-linear / MMSE
Normalized Doppler	0.02 cycles per OFDM symbol
Pilot periods	6, 10, 16, 24 symbols
SNR sweep	0–27 dB in 3 dB steps
Frames per point	100

The reported metrics are

$$\text{BER} = \frac{\text{wrong decoded information bits}}{\text{decoded information bits}}, \quad (5.1)$$

$$\text{EVM}_{\text{rms}} = \sqrt{\frac{\sum_i |\hat{s}_i - s_i|^2}{\sum_i |s_i|^2}}, \quad (5.2)$$

$$\text{NMSE}_H = \frac{\|\hat{\mathbf{H}} - \mathbf{H}\|_F^2}{\|\mathbf{H}\|_F^2}, \quad (5.3)$$

$$\eta_{\text{payload}} = (1 - \text{BER}) \frac{N_{\text{data symbols}}}{N_{\text{total symbols}}}. \quad (5.4)$$

The last metric includes pilot overhead and BER, but excludes FEC-rate overhead, modulation order, and sample rate. It is therefore named normalized payload-symbol efficiency, not absolute spectral efficiency.

Chapter 6

Experiments

This chapter evaluates the effect of pilot density on receiver performance using the developed MIMO-OFDM framework. Unless otherwise stated, all system parameters remain fixed while only the pilot period is varied. Performance is assessed using BER, RMS EVM, channel-estimation NMSE, normalized payload-symbol efficiency, and estimator robustness to Doppler.

6.1 Pilot-Density Reliability

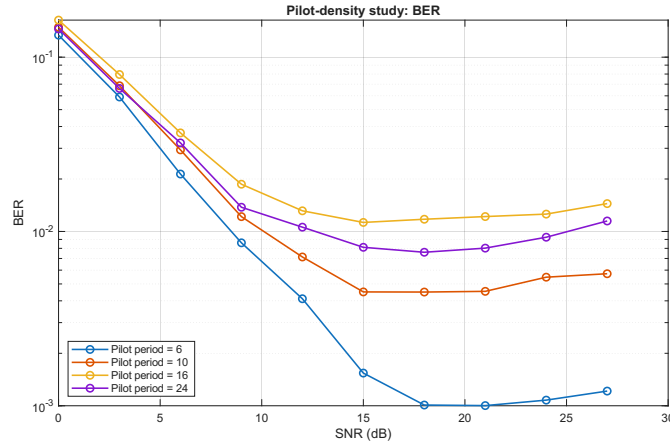


Figure 6.1: BER versus SNR for different pilot periods. Frequent pilot transmission reduces the high-SNR error floor.

Figure 6.1 shows that BER decreases with increasing SNR for all pilot configurations. Frequent pilot transmission ($P = 6$) consistently achieves the lowest BER by providing more recent channel estimates. In contrast, sparse pilot placement results in a high-SNR error floor because interpolation cannot accurately track the time-varying channel, making channel-estimation error the dominant impairment.

6.2 Error-Vector Magnitude

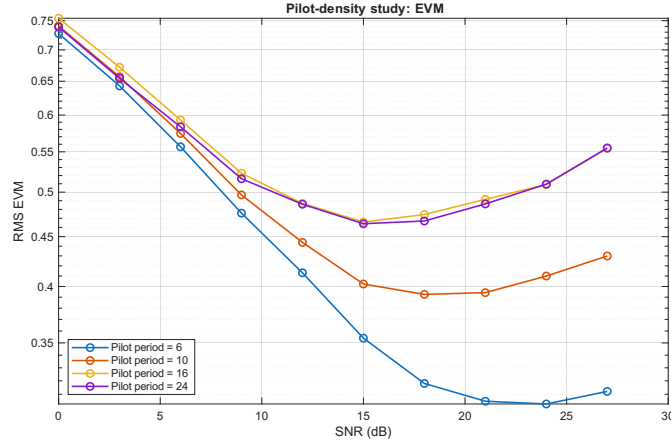


Figure 6.2: RMS EVM versus SNR for different pilot periods. Sparse pilot configurations retain larger residual constellation errors.

Figure 6.2 shows that RMS EVM decreases with increasing SNR before approaching a floor determined by channel-estimation error. Denser pilot transmission maintains lower residual EVM through improved channel tracking, consistent with the BER behaviour observed in Fig. 6.1.

6.3 Channel-Estimation Accuracy

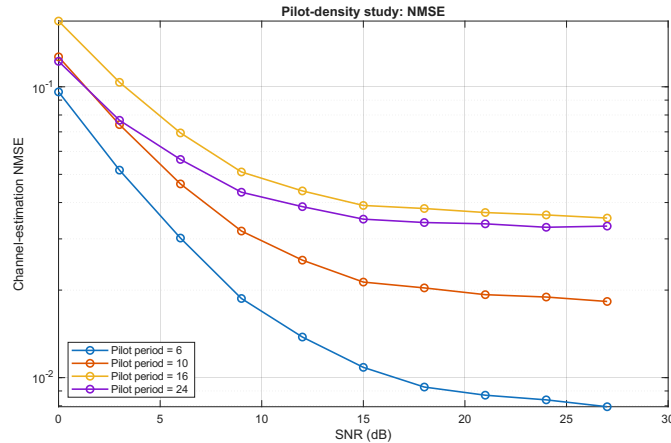


Figure 6.3: Channel-estimation NMSE versus SNR. Frequent pilot updates improve channel tracking accuracy.

Figure 6.3 confirms that denser pilot placement consistently reduces channel-estimation error by providing more frequent channel updates. Sparse pilot configurations rely on increasingly outdated estimates, leading to larger interpolation errors and degraded receiver performance.

6.4 Payload-Efficiency Trade-off

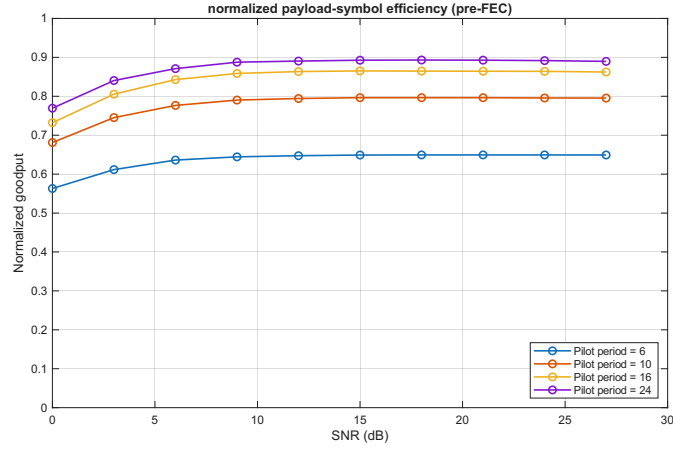


Figure 6.4: Normalized payload-symbol efficiency versus SNR, illustrating the trade-off between pilot overhead and payload transmission.

Increasing pilot density improves channel estimation but also increases pilot overhead. Consequently, sparse pilot configurations achieve higher normalized payload efficiency despite poorer BER performance. The pilot period minimizing BER therefore does not necessarily maximize effective throughput.

6.5 Estimator Robustness to Doppler

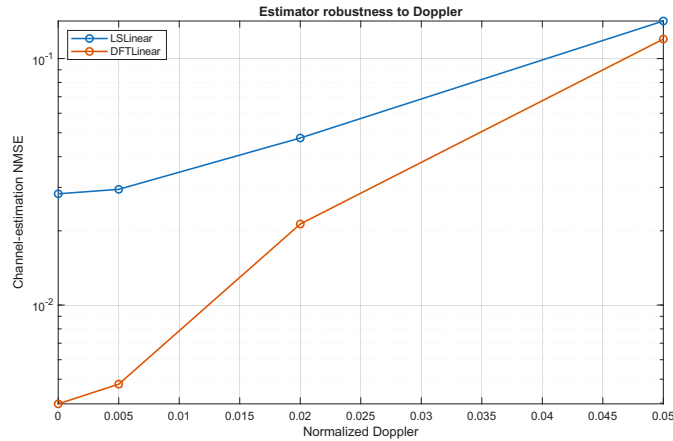


Figure 6.5: NMSE of the LS-linear and DFT-linear estimators versus normalized Doppler frequency.

Figure 6.5 compares estimator performance under increasing Doppler frequency. The DFT-based estimator consistently achieves lower NMSE by exploiting the finite channel delay support. As Doppler increases, both estimators degrade because channel aging becomes dominant, although the DFT-linear estimator retains a clear performance advantage.

Chapter 7

Conclusion

This work developed a reproducible coded MIMO-OFDM research simulator and used it to study pilot-density tradeoffs. The results show a clear engineering balance: frequent pilots improve channel estimates, EVM, and BER in time-varying multipath channels, while sparse pilots allocate more OFDM symbols to data. DFT-domain delay truncation improves estimation when delay support is known. The system is a sound generic baseband research platform, and the documented limitations define a transparent path toward standards-oriented extensions.

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